

Experience

Research Software Engineer, Institute of Cancer Research

October 2024 – Present

- Partnered directly with researchers to gather requirements and develop software solutions supporting cancer research, ranging from data-processing pipelines to full-stack Flask/MySQL web applications.
- Built and maintained deployment infrastructure for internal research applications using Docker, NGINX, MySQL, and CI/CD automation.
- Optimised computational performance of large-scale workflows, reducing HPC resource usage and improving reproducibility for publication-ready analyses.
- Delivered Carpentries-style training and provided ongoing support in software engineering best practices, code optimisation, and effective use of HPC environments.
- Created written and video tutorials (YouTube) to help researchers run software on the Alma HPC cluster at ICR.

Technical Skills: *Python (Flask, FastAPI), R, C++, Julia, Bash, SQL; Git, CI/CD (GitLab runners, automated testing/deployment); HPC, SLURM, Docker, Linux, Virtual Machines; Data Analysis, Machine Learning (VAEs, GANs); Data Structures & Algorithms.*

Academic Papers

Designing minimal E. coli genomes using variational autoencoders (2024)

First author — Submitted to *Cell Systems*

Generative design and construction of functional plasmids with a DNA language model (2025)

Co-author — Submitted to *Nature Biotechnology*

Education

MSc Computer Science, University College London, UK

September 2023 – September 2024

Modules: Introduction to Programming | Database Fundamentals | Algorithms | Computer Architecture | App Engineering | Software Engineering | Functional Programming | Applied Deep Learning | Computer Science Project | Overall: Merit

BSc Biomedical Sciences: Cells and Molecules, University College London, UK

September 2020 – May 2023

Relevant Modules: Introductory Statistical Methods and Computing | Introduction to Coding for Biosciences Research | Linear Algebra for Data Science | Advanced Statistics and Machine Learning for Biosciences | Overall: First Class

Academic Experience / Extra Education

Master's Thesis at Prof. Chris Barnes' Lab

June 2024 – September 2024

Title: *Minimal Bacterial Genomes using Generative Machine Learning Model*. Working with collaborators at the University of Bristol, built a variational autoencoder (VAE) to generate reduced *E. coli* genomes, stripped to only essential genes. Outputs were validated using Bristol's whole-cell model. The project resulted in a first-author preprint submitted to *Cell Systems*. The generative approach developed here directly opened a follow-on direction in plasmid design, subsequently developed into a separate preprint submitted to *Nature Biotechnology*.

Summer Project at Dr. Charles Swanton's Lab

June – July 2023

Studied ways of applying autoencoders on genetic data for cancer classification and investigated VAE models on gene expression data. Additionally, pre-processed and organised a 300-page document with cancer gene information into a data frame for use in future diagnostic pipelines.

Bachelor's Thesis at Prof. Chris Barnes' Lab

September 2022 – March 2023

Title: *Exploration of Variant Importance in Cancer Diagnostics using Machine Learning*. Built a self-supervised machine learning workflow for predicting cancer type from genomic data and analysed the impact of SNVs, CNVs, and SVs on prediction accuracy using gradient-boosted trees with CatBoost.

Summer Project at Prof. Richard Mott's Lab

May – September 2022

Spent four months analysing rat RNA sequencing gene expression data for signatures of dominance in alternative splicing across Liver and Fat tissues, building bioinformatics analysis pipelines from scratch – an experience that motivated a longer-term interest in building more intelligent, automated systems for scientific discovery.

References

Prof. Chris Barnes

University College London

✉ christopher.barnes@ucl.ac.uk

Rachel Alcraft

Institute of Cancer Research

✉ rachel.alcraft@icr.ac.uk